

BALWA MOHAMMADALI

Machine Learning Engineer

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EDUCATION

Bachelor of Engineering (B.E.) in Computer Engineering

2023 – Present

Gujarat Technological University, Ahmedabad

CGPA: 9.26 / 10

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Scikit-Learn, Django, FastAPI, Pydantic, Tensorflow/keras

ML/DL: Supervised Learning, Unsupervised Learning, Natural Language Processing (NLP), ANN, CNN, RNN, LSTM

Tools & Platforms: Git/GitHub, VS Code, Jupyter Notebook, Google Colab, MySQL, Excel, n8n

EXPERIENCE

Generative AI Intern – SmartBridge | Remote

3 jul 2026 – 17 Jul 2026

- Completed the GenAI Career Accelerator Program under NASSCOM FutureSkills Prime, covering Generative AI, LLMs, prompt engineering, and AI agents.
- Gained hands-on exposure to vibe coding and data acquisition for AI-powered applications.
- Built a Generative AI web application as a capstone project using the Antigravity IDE.
- Applied GenAI concepts and prompt engineering through an AI-assisted development workflow.

Machine Learning Intern – Edunet Foundation | Remote

13 May 2025 – 13 Jun 2025

- Developed an end-to-end Heart Disease Prediction system using Machine Learning and healthcare datasets for predictive analysis.
- Built and implemented an ML pipeline including data preprocessing, feature engineering, model training, evaluation, and prediction.
- Performed exploratory data analysis (EDA) and data visualization using Pandas and NumPy to extract healthcare insights.
- Trained and evaluated classification models using Scikit-Learn and compared performance using standard evaluation metrics.

PROJECTS

Plant Disease Detection System | EfficientNetB0, TensorFlow/Keras, FastAPI, Grad-CAM

- Built a deep learning-based plant disease classification system using EfficientNetB0 transfer learning to classify images into 39 plant health and disease categories.
- Applied image preprocessing and augmentation and fine-tuned the final 20 layers of EfficientNetB0, achieving 99.15
- Evaluated model performance using precision, recall, and F1-score, achieving 99
- Implemented Grad-CAM for visual model explainability and integrated the trained model with a FastAPI prediction API for image-based inference.

Image-Based Cattle & Buffalo Breed Recognition | MobileNet, TensorFlow/Keras, Django

- Developed a deep learning-based cattle and buffalo breed classification system using MobileNet transfer learning for image-based breed recognition.
- Applied image preprocessing and data augmentation to improve model generalization and classification performance.
- Achieved approximately 74
- Integrated the trained model into a Django web application with image upload, real-time prediction, Top-3 breed results, and a responsive HTML/CSS/JavaScript interface.

Movie Recommendation System | NLP, Scikit-Learn, FastAPI

- Built a content-based movie recommendation system using movie metadata and NLP techniques.
- Performed data preprocessing and feature engineering by combining genres, keywords, cast, crew, and movie overview into a unified feature set.
- Applied CountVectorizer to convert textual features into numerical vectors and used cosine similarity to generate movie recommendations.
- Integrated TMDB API to fetch movie posters and display recommendations through an interactive web interface.

RAG-Based AI Agent for PYQ Analysis | RAG, Google Gemini, Pinecone, AI Agents, n8n

- Built an end-to-end RAG-based AI agent to help students retrieve and analyze Previous Year Questions (PYQs) in an exam-oriented manner.
- Automated Google Drive document ingestion and applied Recursive Character Text Splitting to organize academic content into retrievable chunks.
- Generated semantic embeddings using Google Gemini and stored them in Pinecone Vector Store for efficient similarity-based retrieval.
- Integrated an AI Agent with retrieval-augmented generation to answer student queries using relevant PYQ context and provide structured, context-grounded responses.

IPL Win Probability Prediction | Python, Scikit-Learn, FastAPI

- Developed a machine learning model to predict real-time IPL match win probability using historical ball-by-ball and match datasets.
- Performed data preprocessing, cleaning, and feature engineering using runs left, balls left, wickets left, CRR, and RRR.
- Applied One-Hot Encoding to transform categorical features including teams and match locations into model-ready inputs.
- Trained and optimized a Logistic Regression model using GridSearchCV, achieving approximately 93% prediction accuracy.

KitabKhana – Online Book Store | React, PHP, MySQL

- Developed a full-stack online book store enabling users to Buy, Rent, and Donate books through a responsive React-based interface.
- Built RESTful APIs using PHP and MySQL to manage authentication, book listings, inventory, and order processing.
- Implemented core marketplace features including rental management, wishlist functionality, customer reviews, and order tracking.
- Designed a modern UI using React, Vite, and Tailwind CSS with responsive layouts and dark/light mode support.

HACKATHONS & COMPETITIONS

- Smart India Hackathon 2025 – Developed a CNN-based machine learning solution for cattle and buffalo breed recognition using computer vision techniques.
- Odoo Hackathon 2025 – Designed and developed a sustainable clothing exchange platform featuring point-based redemption and inventory management workflows.
- UNESCO Global Youth Hackathon 2025 – Built an NLP and machine learning-based fake news detection system for real-time text classification and prediction.
- SSIP VikaSaptah Hackathon – Developed a consumer complaint tracking system with complaint management, live status tracking, and field operation workflows.

CERTIFICATIONS

- Python – HackerRank
- Machine Learning – IBM SkillsBuild
- Java Programming – Great Learning